TypeScript

TS

Utility Types

Pick

Select specific properties from an existing type

```
type User = {
  name: string;
  age: number;
  email: string;
};

type UserPreview = Pick<User, "name" | "email">;

const user: UserPreview = {
  name: "Munna",
  email: "munna@example.com"
};
```

Why use it?

Create smaller types from large objects

In Short

Pick = choose only what you need

Omit

Remove specific properties from an existing type

```
type User = {
  name: string;
  age: number;
  email: string;
};

type UserWithoutEmail = Omit<User, "email">;

const user: UserWithoutEmail = {
  name: "Munna",
  age: 25
};
```

Why use it?

Exclude fields you don't need

In short

Omit = remove what you don't need

Required

Make all properties of a type required

```
type User = {
  name: string;
  age?: number;
  email?: string;
};

type FullUser = Required<User>;

const user: FullUser = {
  name: "Munna",
  age: 25,
  email: "munna@example.com"
};
```

Why use it?

Ensure all optional fields must be provided

In short

Required = no optional properties

Partial

Make all properties of a type optional

```
type User = {
  name: string;
  age: number;
  email: string;
};

type UpdateUser = Partial<User>;

const user: UpdateUser = {
  name: "Munna"
};
```

Why use it?

Useful for updates where not all fields are required

In short

Partial = everything becomes optional

Readonly

Make all properties of a type read-only

```
type User = {
  name: string;
  age: number;
};

const user: Readonly<User> = {
  name: "Munna",
  age: 25
};

// user.age = 30 ❌ Error (cannot modify)
```

Why use it?

Prevent accidental changes to data

In short

Readonly = values can't be changed

Record

Create an object type with specific keys and value types

```
type UserRoles = Record<string, string>;

const roles: UserRoles = {
  admin: "full access",
  user: "limited access"
};
```

What is happening?

- Keys → defined by the first type
- Values → defined by the second type

In short

Record = define object shape with key-value pairs

Thank You

Follow for more